

# Robot Learning: From Fundamentals to Foundation Models

Lecture 2: Robot Control & Markov Decision Processes

Oier Mees

**ETH** zürich



Microsoft

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23.02.2026

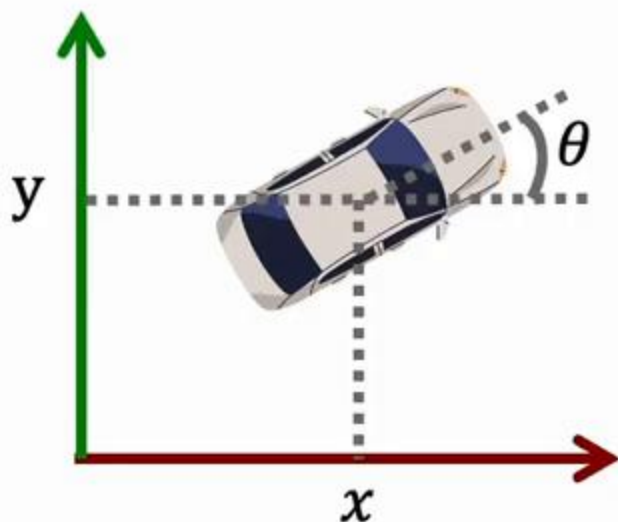
## Feedback → Actions

- Extended deadline for 1st homework to March 5th
- Created a moodle forum for QA
- Linked Course Website in MyStudies
- Made slides & recordings available
  - login:
  - pwd:
- Instructions for Paper Discussion

# Robot Motion

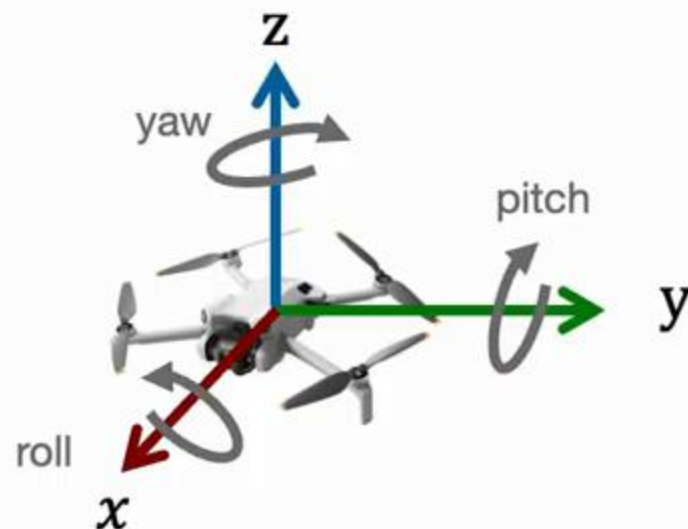
- Can be represented as **rigid-body motion**

**Planar Motion**



SE(2): The special Euclidean group of rigid body motion in 2D

**Spatial Motion**



SE(3): The special Euclidean group of rigid body motion in 3D

# Mathematical Representation

## SE(2)

3 “Degrees of Freedom”

$$T_{SE(2)} = \begin{bmatrix} \cos \theta & -\sin \theta & x \\ \sin \theta & \cos \theta & y \\ 0 & 0 & 1 \end{bmatrix}$$

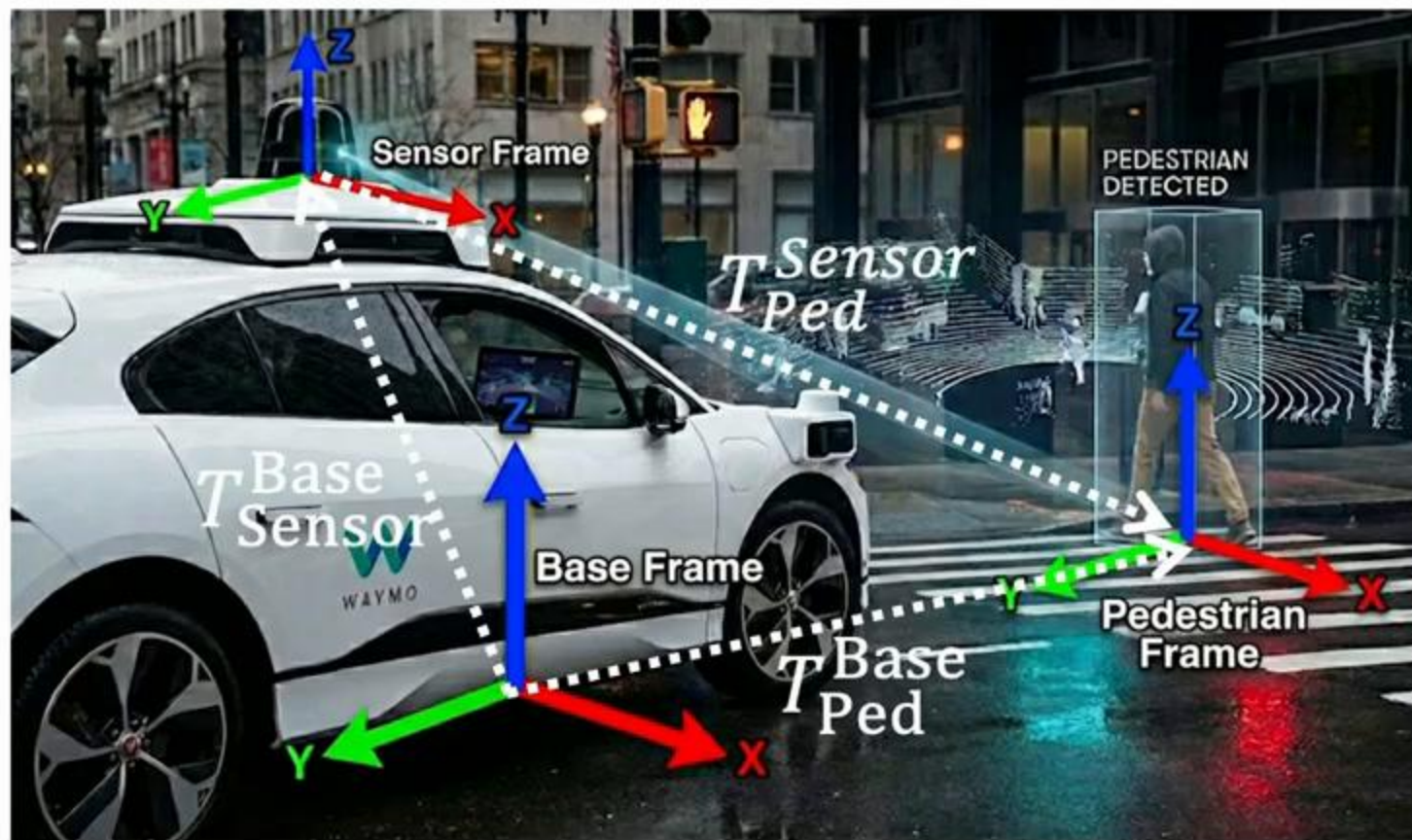
## SE(3)

6 “Degrees of Freedom”

$$T_{SE(3)} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & x \\ r_{21} & r_{22} & r_{23} & y \\ r_{31} & r_{32} & r_{33} & z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$SO(n) = \{R \in \mathbb{R}^{n \times n} \mid R^T R = I, \det(R) = 1\}$$

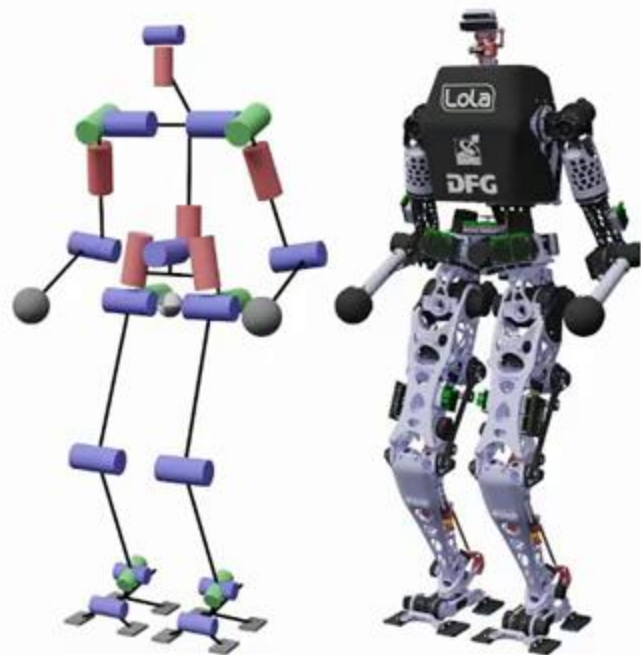
# Chaining Transforms



$$T_{Ped}^{Base} = T_{Sensor}^{Base} \times T_{Ped}^{Sensor}$$

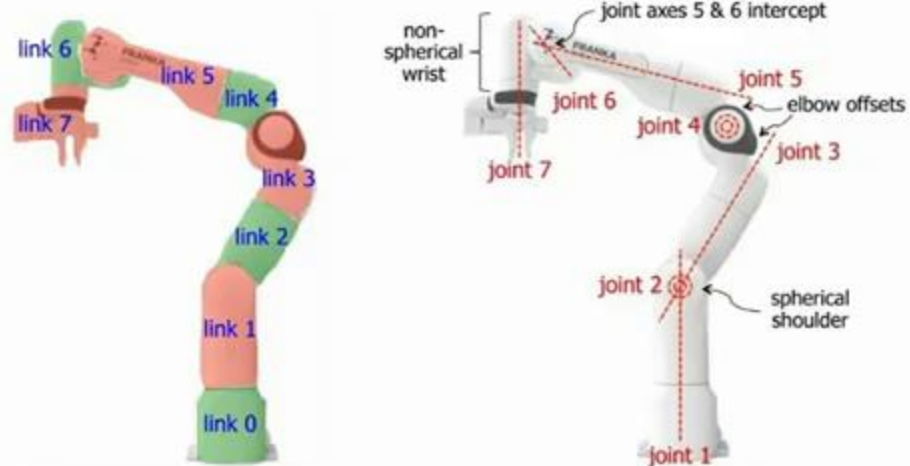
# Articulated Robot Bodies

- Robots can be defined as articulated rigid bodies, consisting of:
  - **Link:** a single rigid body
  - **Joint:** connection between links
  - **End-Effector:** a device attached to a specific link, like end of arms

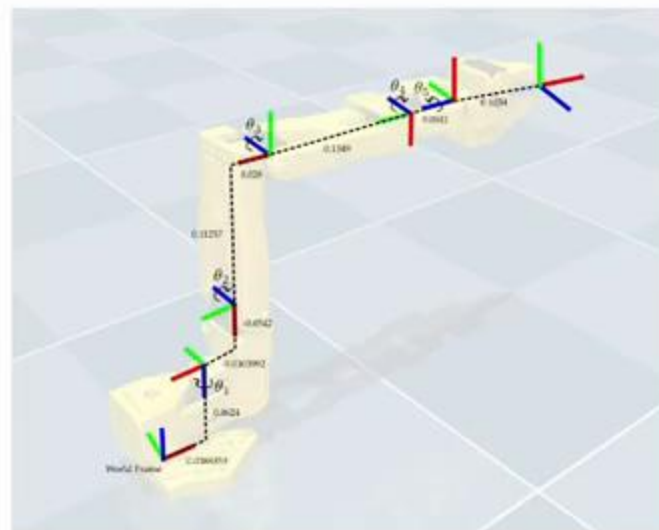


LOLA Humanoid Robot @ TUM

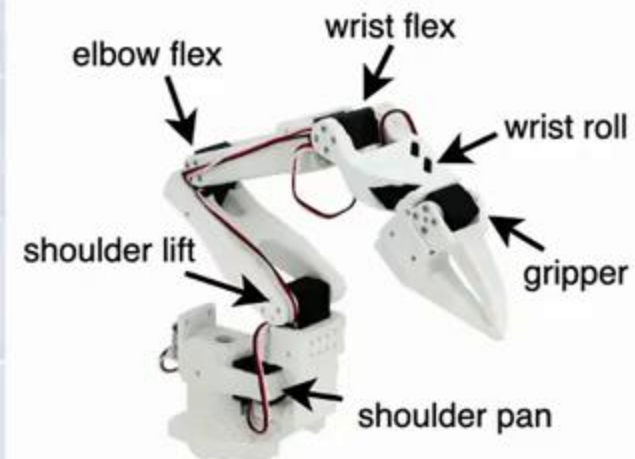
# Robot Arm Articulations



Franka Emika



SO-101



# Robot End Effectors



Suction Gripper



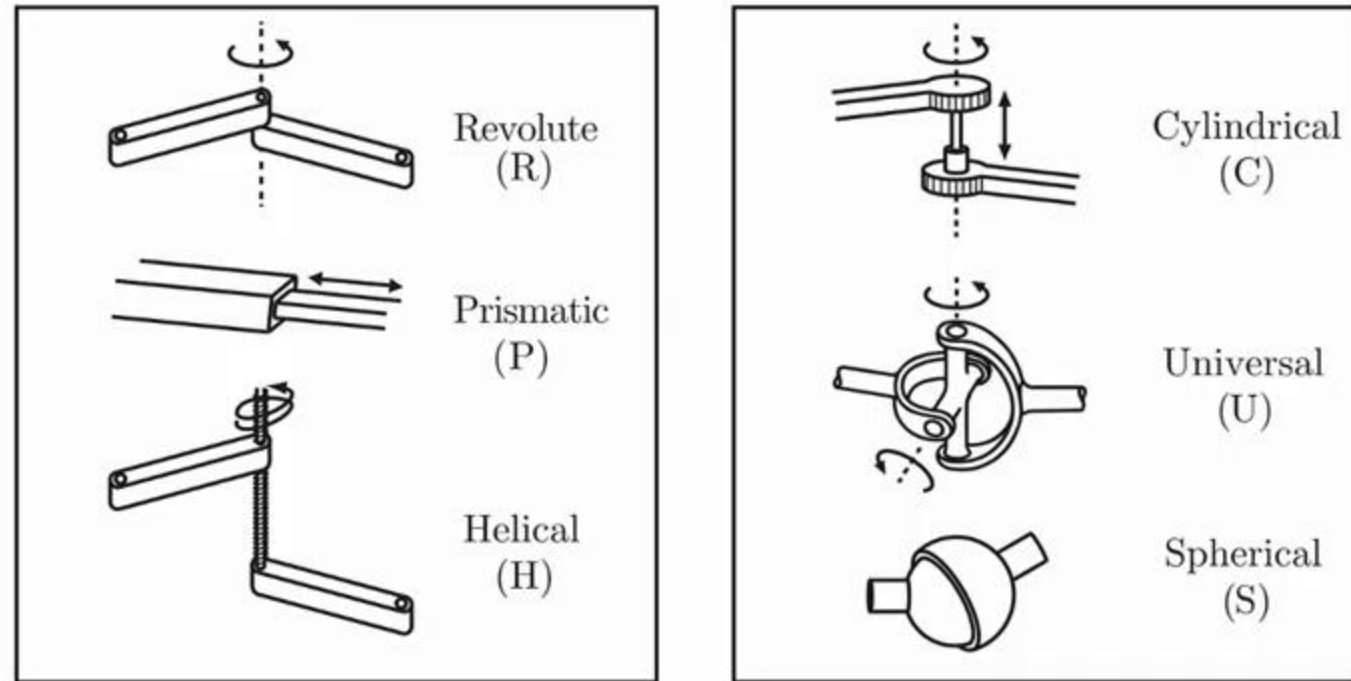
Parallel Gripper



Dexterous Hand

# Robot Joints

- A joint provides constraints on the possible motions of the two rigid bodies it connects.



Source: Modern Robotics by K. Lynch 2017

# Degrees of Freedom

- Number of independent parameters required to completely specify the configuration or state of the robot

| Joint type      | dof $f$ | Constraints $c$<br>between two<br>planar<br>rigid bodies | Constraints $c$<br>between two<br>spatial<br>rigid bodies |
|-----------------|---------|----------------------------------------------------------|-----------------------------------------------------------|
| Revolute (R)    | 1       | 2                                                        | 5                                                         |
| Prismatic (P)   | 1       | 2                                                        | 5                                                         |
| Helical (H)     | 1       | N/A                                                      | 5                                                         |
| Cylindrical (C) | 2       | N/A                                                      | 4                                                         |
| Universal (U)   | 2       | N/A                                                      | 4                                                         |
| Spherical (S)   | 3       | N/A                                                      | 3                                                         |

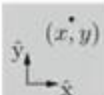
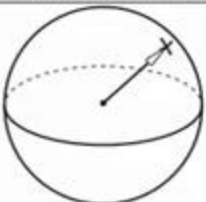
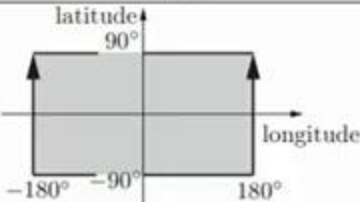
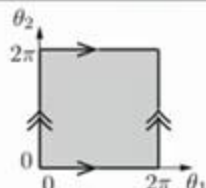
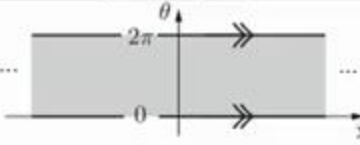
Chebychev–Grübler–Kutzbach Formula:

$$\text{dof} = \underbrace{m(N-1)}_{\text{rigid body freedoms}} - \underbrace{\sum_{i=1}^J c_i}_{\text{joint constraints}}$$

Source: Modern Robotics by K. Lynch 2017

# Configuration Space

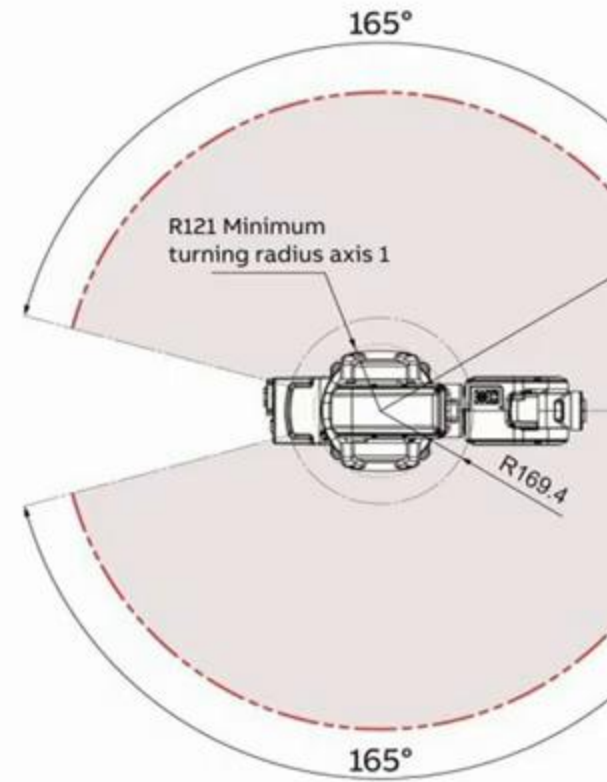
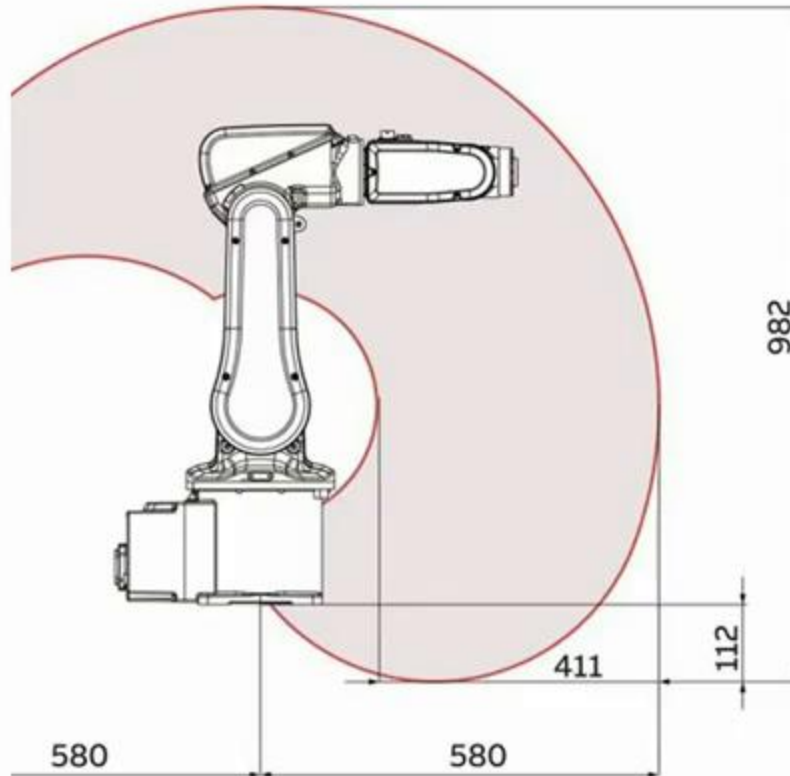
- Set of all possible robot configurations

| system                                                                                                        | topology                                                                                                           | sample representation                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <br>point on a plane        | <br>$\mathbb{E}^2$              | <br>$\mathbb{R}^2$                                         |
| <br>spherical pendulum      | <br>$S^2$                       | <br>$[-180^\circ, 180^\circ] \times [-90^\circ, 90^\circ]$ |
| <br>2R robot arm           | <br>$T^2 = S^1 \times S^1$      | <br>$[0, 2\pi) \times [0, 2\pi)$                          |
| <br>rotating sliding knob | <br>$\mathbb{E}^1 \times S^1$ | <br>$\mathbb{R}^1 \times [0, 2\pi)$                      |

Source: Modern Robotics by K. Lynch 2017

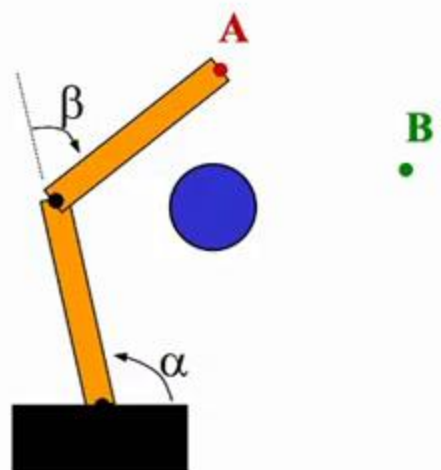
# Workspace

- All points in space reachable by the robot's end-effector

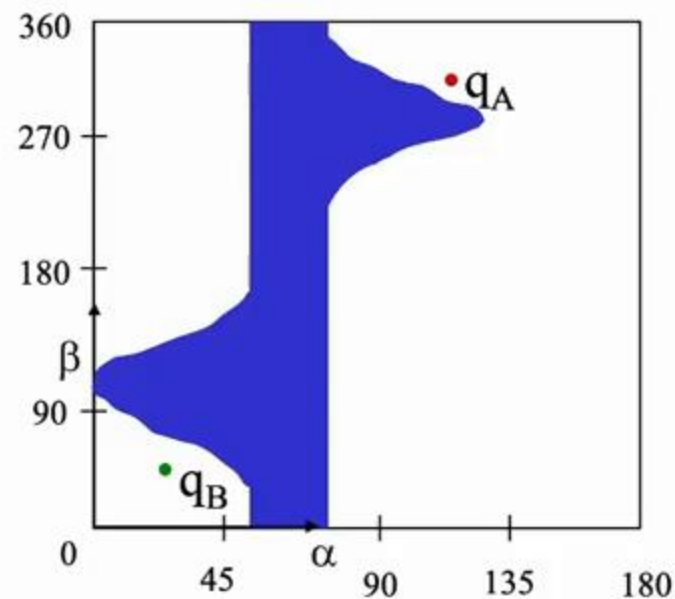


Source: ABB, Technical Data for The IRB 120 Industrial Robot

# Obstacles in Configuration Space



An obstacle in the robot's workspace

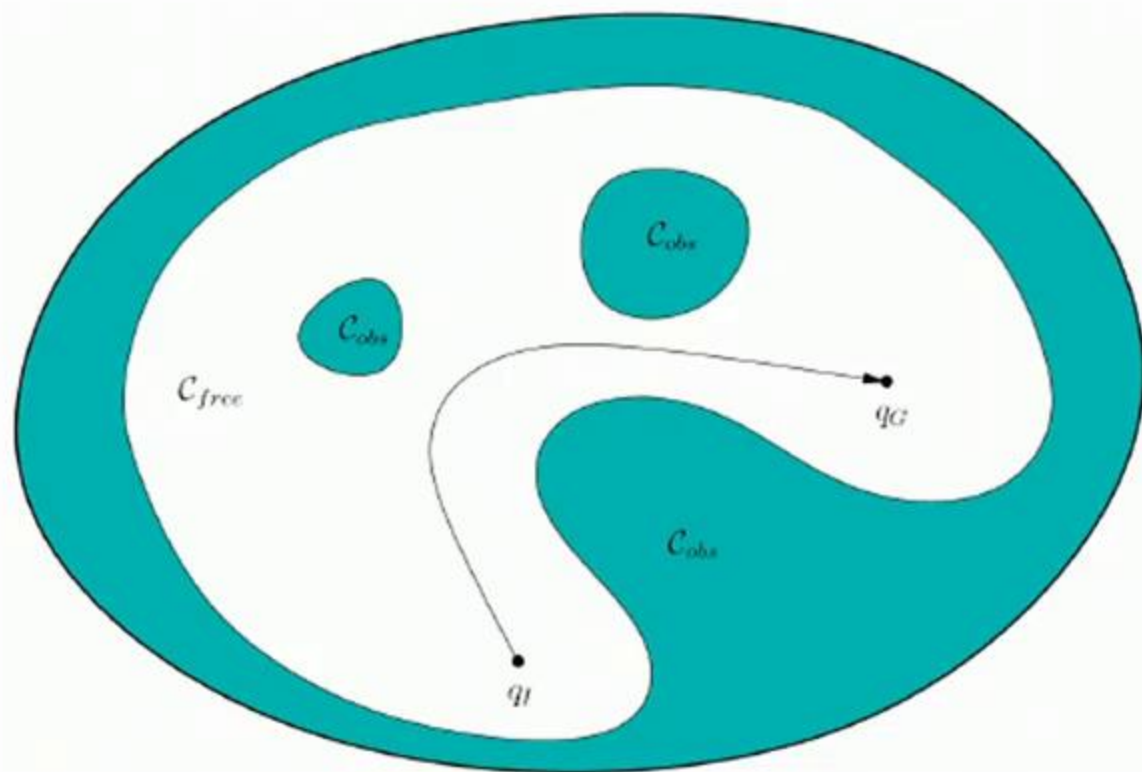


Configuration Space representation of this obstacle

Source: Robot Motion Planning by JC Latombe 1991

# Motion Planning in C-Space

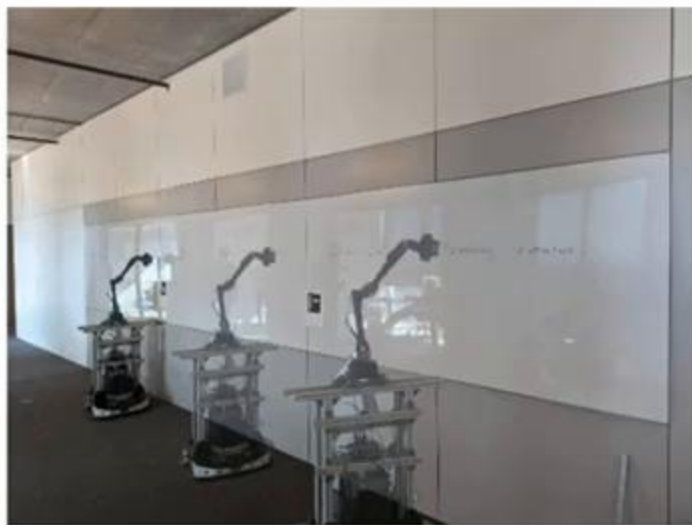
- C-Space used for robot motion planning
- Decouples robot geometry from path planning



Source: Planning Algorithms by Steven M. LaValle 2006

# Task Space

- The manifold in which the robot's task is naturally defined, independent of its embodiment
- If Task Space dimension < Robot's DoF = Redundancy



Task: whiteboard cleaning  
Task Space: whiteboard surface  $\mathbb{R}^2$



Task: clean room floor  
Task Space: vacuum robot's position on the floor  $\mathbb{R}^2$

# Question Time!

## Configuration Space?

7D Manifold ( $\theta_1 \dots \theta_7$ )

## Workspace?

Reachable physical  
space ( $x, y, z$ )

## Task Space?

6D Pose of end-effector SE(3)



# Question Time!

## ■ Redundancy → Null Space Motion

**Configuration Space?**

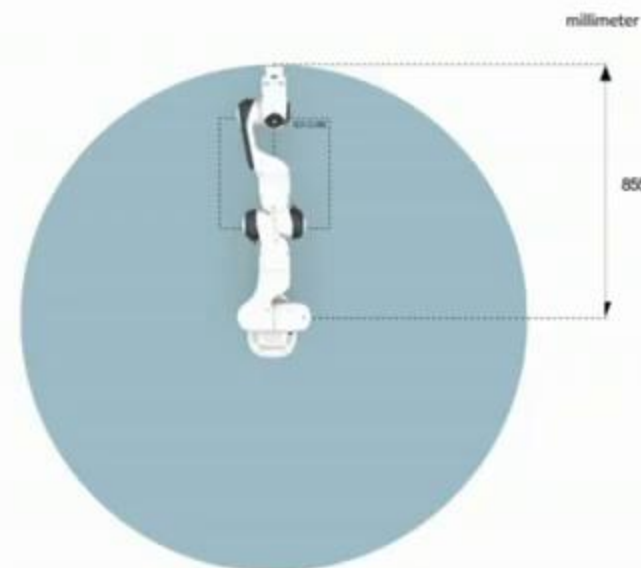
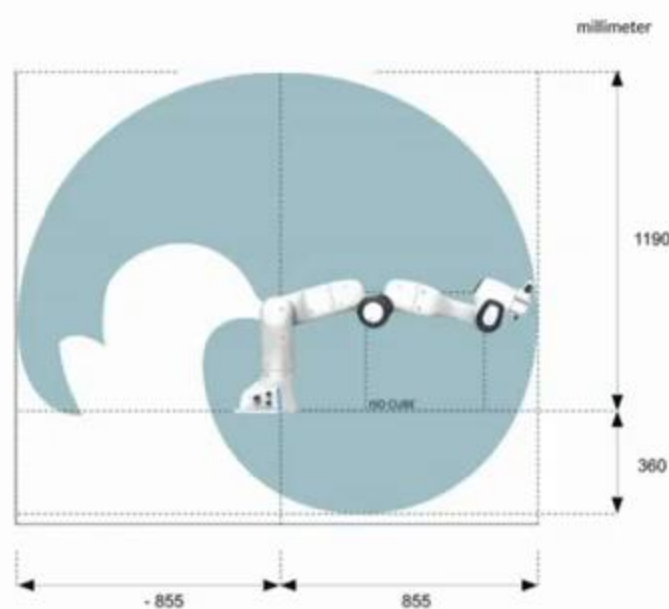
7D Manifold ( $\theta_1 \dots \theta_7$ )

**Workspace?**

Reachable physical  
space ( $x, y, z$ )

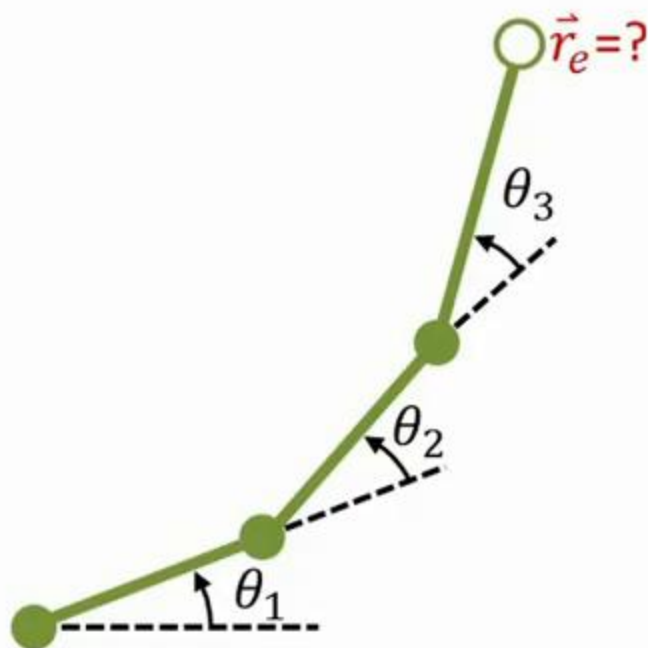
**Task Space?**

6D Pose of end-effector SE(3)



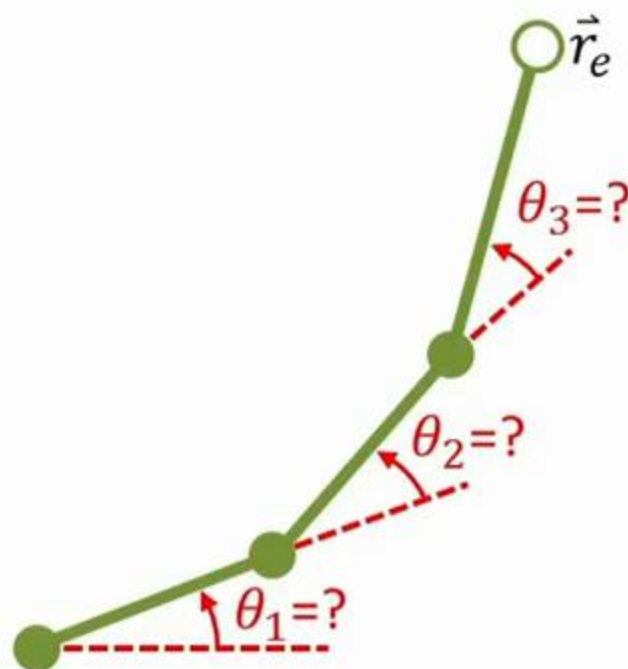
# Forward Kinematics

- Given robot joints, what is the end-effector pose?
- Mapping from  $C$  to Task Space,  $f: C \rightarrow X$
- $x=f(q)$ ,  $q \in \mathbb{R}^n$ ,  $x \in SE(3)$  deterministic, but not bijective



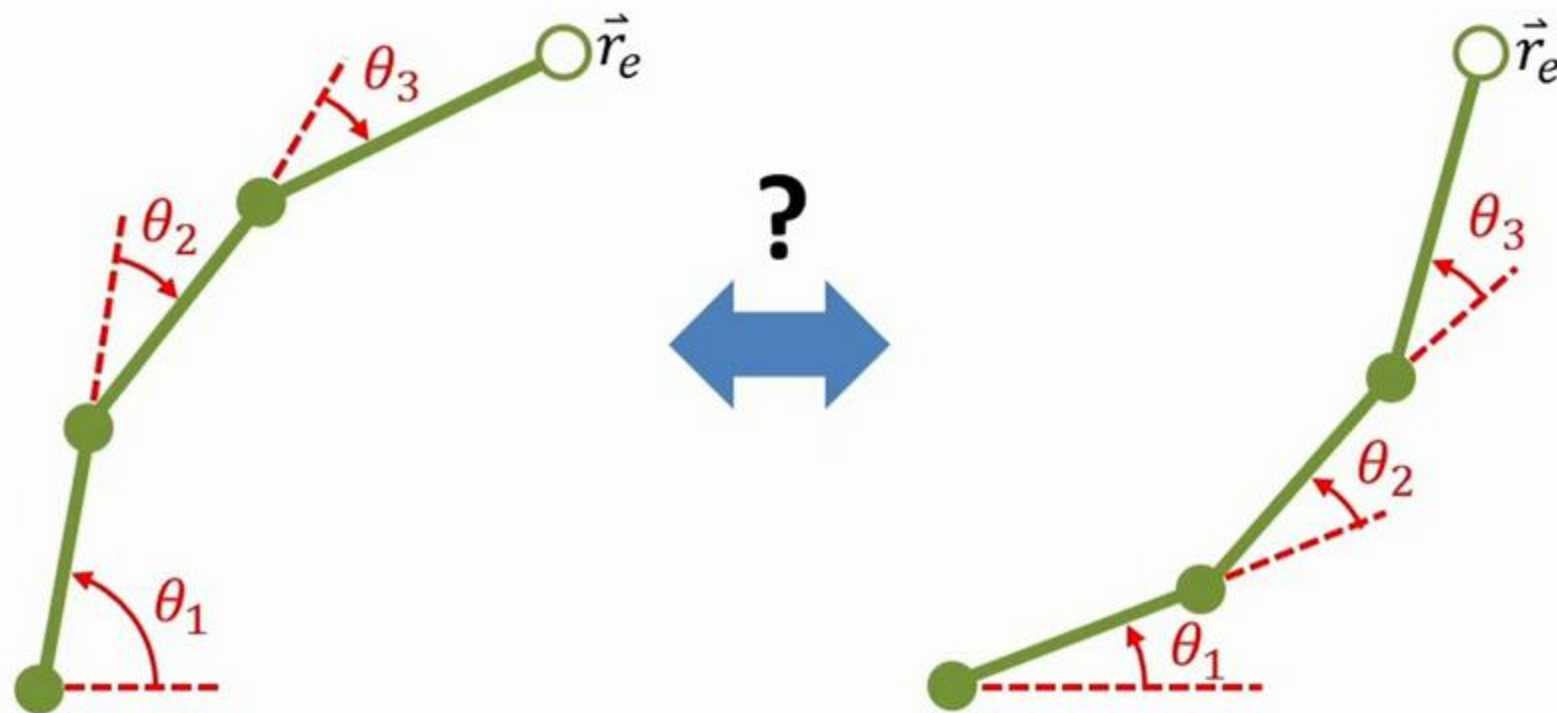
# Inverse Kinematics

- Given end-effector pose, what are the joint configurations?
- $q = f^{-1}(x)$ , mapping  $f^{-1}: \mathcal{X} \rightarrow \mathcal{P}(\mathcal{C})$



# Inverse Kinematics

- Often **non-unique** or has no analytical solution
- Optimization-based IKs



# Optimization-Based IK

- **Objective:** Minimize the distance between the current end-effector and the target

- **Loss:**  $L(\theta) = \frac{1}{2} || \overbrace{f(\theta) - x_{\text{target}}}^{\mathbf{e}} ||^2$

- **Update:**  $\theta_{\text{new}} = \theta_{\text{old}} - \alpha \nabla_{\theta} L(\theta)$

- **Gradient of the Loss:**  $\nabla_{\theta} L(\theta) = \left( \frac{\partial f}{\partial \theta} \right)^T \mathbf{e}$

- **Jacobian Method:**  $\theta_{\text{new}} = \theta_{\text{old}} - \alpha J(\theta)^T (f(\theta) - x_{\text{target}})$

# Converting IK to Motor Commands

- IK gives the target joints, but robot can't "jump" there
- Time for movement?



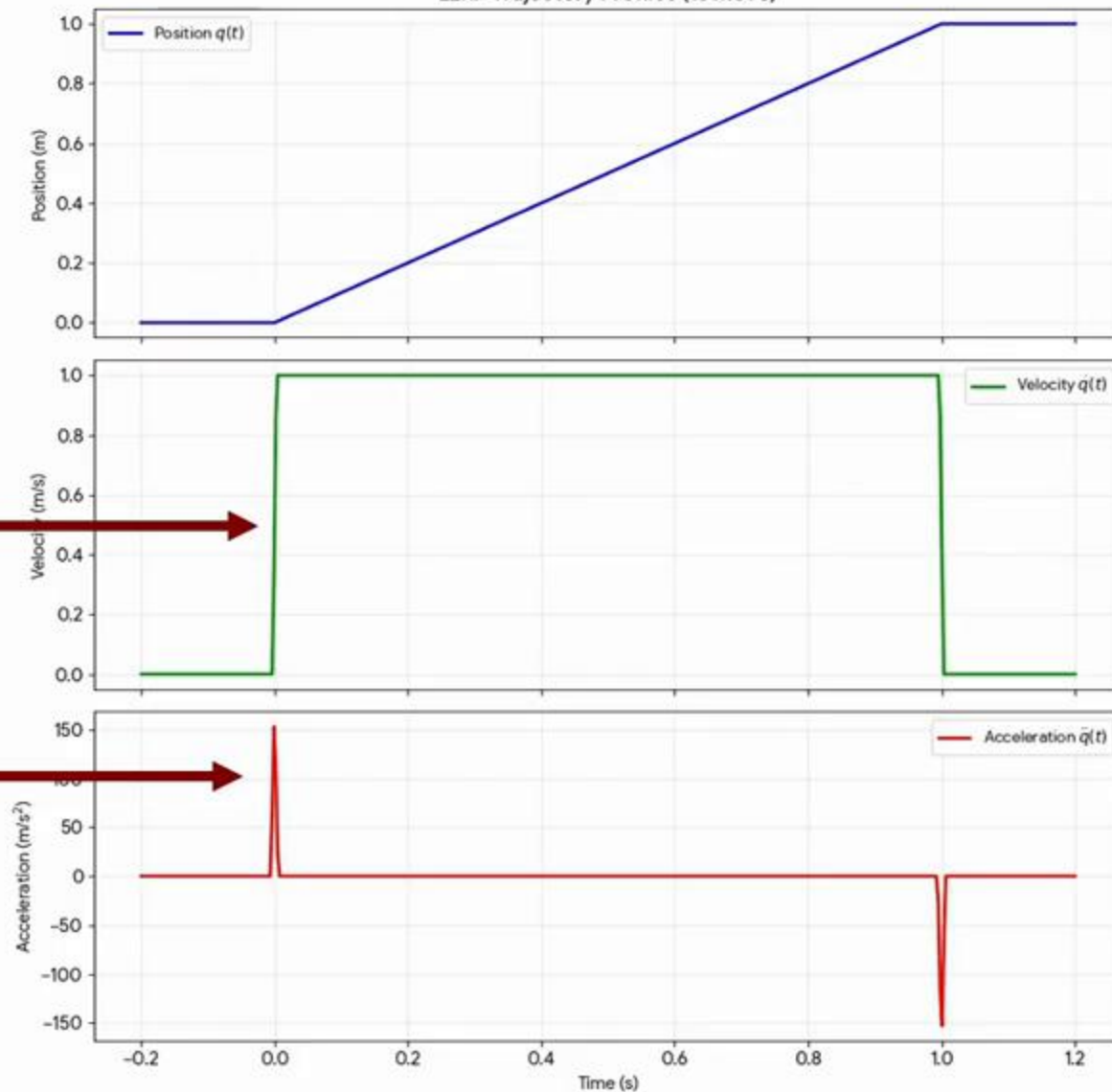
# Trajectory Waypoint Generation: LERP

- Inputs:  $q_{\text{start}}, q_{\text{target}} \in R^6$ , time  $T$ , control frequency  $f$
- Number of Waypoints  $N = \lceil T \cdot f \rceil$
- Normalized Time Progress  $s_i = \frac{i}{N}, s_i \in [0,1]$
- Linear Interpolation  $q(s_i) = q_{\text{start}} + s_i \cdot (q_{\text{target}} - q_{\text{start}})$



# Issues with LERP

LERP Trajectory Profiles (1s Move)



**Instant Max Speed!**  
Real motors can't reach  
full speed in zero time

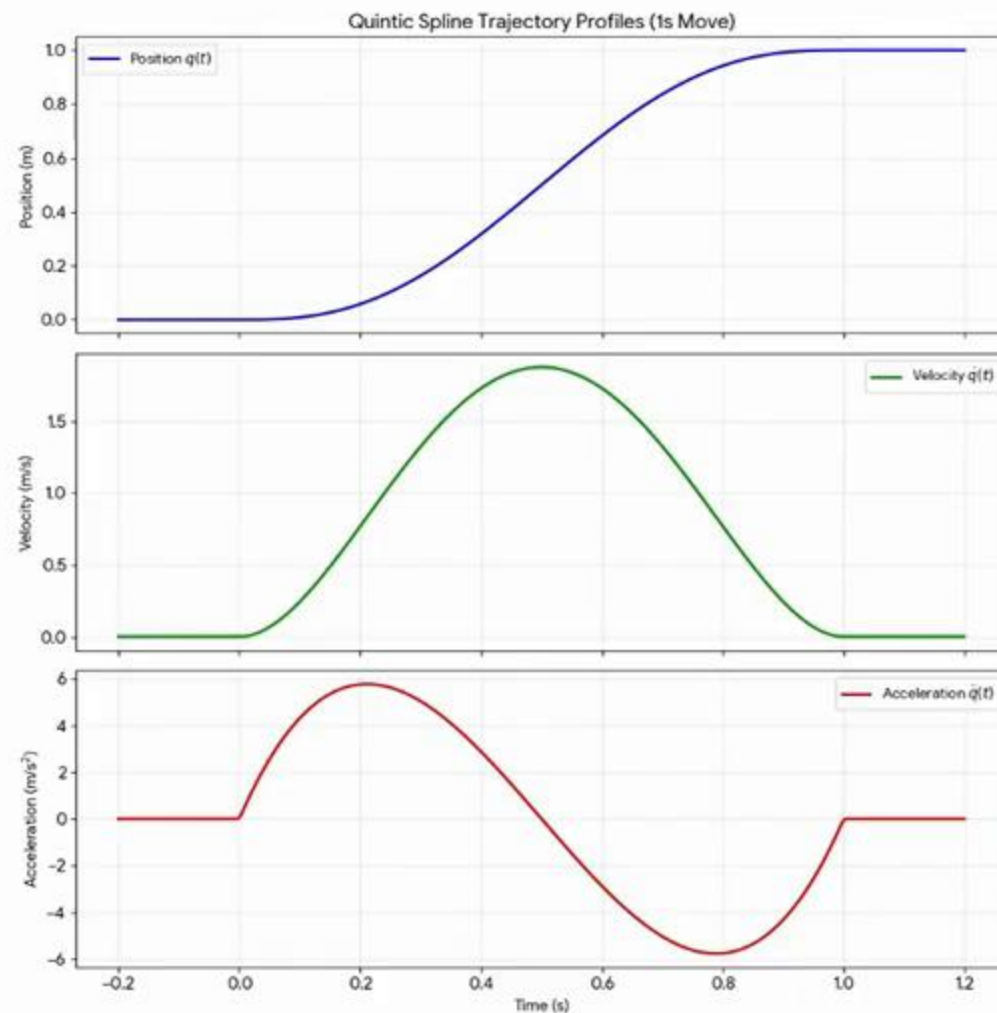
**High Jerk, Dirac Deltas**  
Stress on actuators, vibration

# Trajectory Waypoint Generation: Quintic Splines

- $q(t) = a_0 + a_1t + a_2t^2 + a_3t^3 + a_4t^4 + a_5t^5$
- 6x Boundary Conditions for Smoothness:
  - Start & End with no velocity and accel. while keeping position
- $q(s_i) = q_{\text{start}} + (q_{\text{target}} - q_{\text{start}}) \cdot f(s_i) \quad s_i \in [0,1]$
- Quintic Time Scaling  $f(s_i) = 10s_i^3 - 15s_i^4 + 6s_i^5$

# Trajectory Waypoint Generation: Quintic Splines

- Standard for industry robots like Frankas



# PID Control

- Given trajectory waypoints  $q(s_i)$  and current measurement, continuously calculate corrective motor commands ( $u_k$ )
- Tracking error  $e(t) = q_{desired}(t) - q_{measured}(t)$
- Continuous form  $u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$
- Discrete form  $u_k = \underbrace{K_p e_k}_{\text{Proportional}} + \underbrace{K_i \sum_{j=0}^k (e_j \cdot \Delta t)}_{\text{Integral}} + \underbrace{K_d \frac{e_k - e_{k-1}}{\Delta t}}_{\text{Derivative}}$

# PID Control

$$u_k = \underbrace{K_p e_k}_{\text{Proportional}} + \underbrace{K_i \sum_{j=0}^k (e_j \cdot \Delta t)}_{\text{Integral}} + \underbrace{K_d \frac{e_k - e_{k-1}}{\Delta t}}_{\text{Derivative}}$$

**Proportional**

**“Spring”**

The bigger the tracking error,  
the harder the robot pulls

**Integral**

**“Anti-Gravity”**

Helps compensate if  
gravity is too much

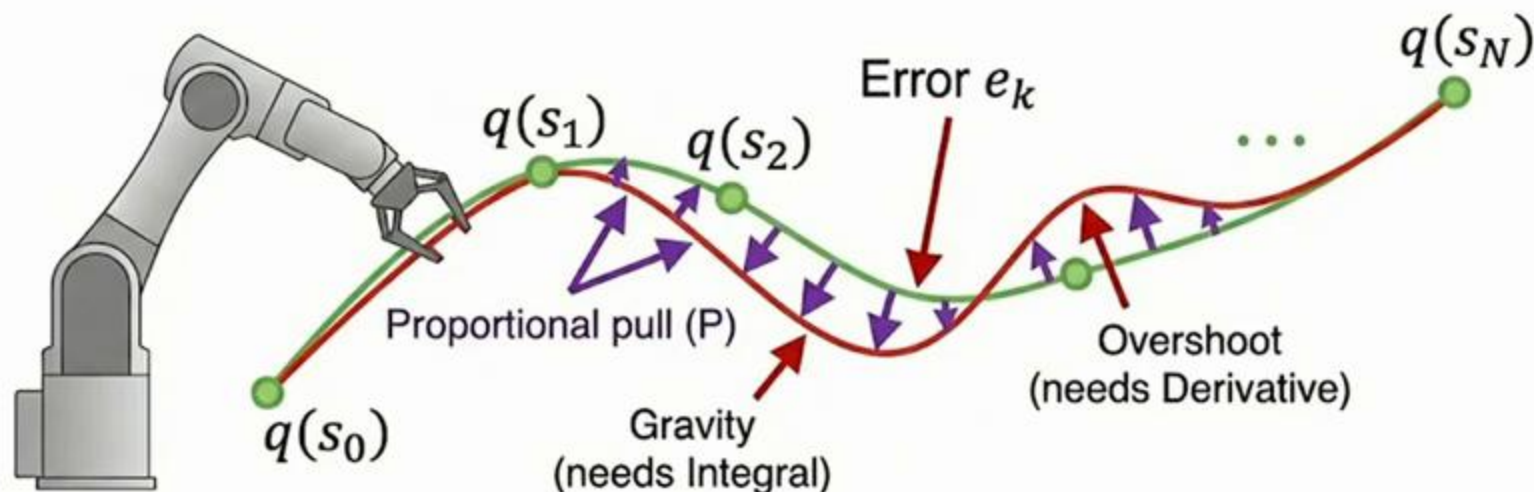
**Derivative**

**“Damper”**

Stops the arm from  
overshooting past  
a waypoint

$$\Delta t = 1/f$$

If frequency too low,  
“Damper” might not be  
able to react fast  
enough



**So we know how to plan & execute robot motion  
are we done?**

Not quite...

# Robot Learning

Solving **Robotics** via **Machine Learning**

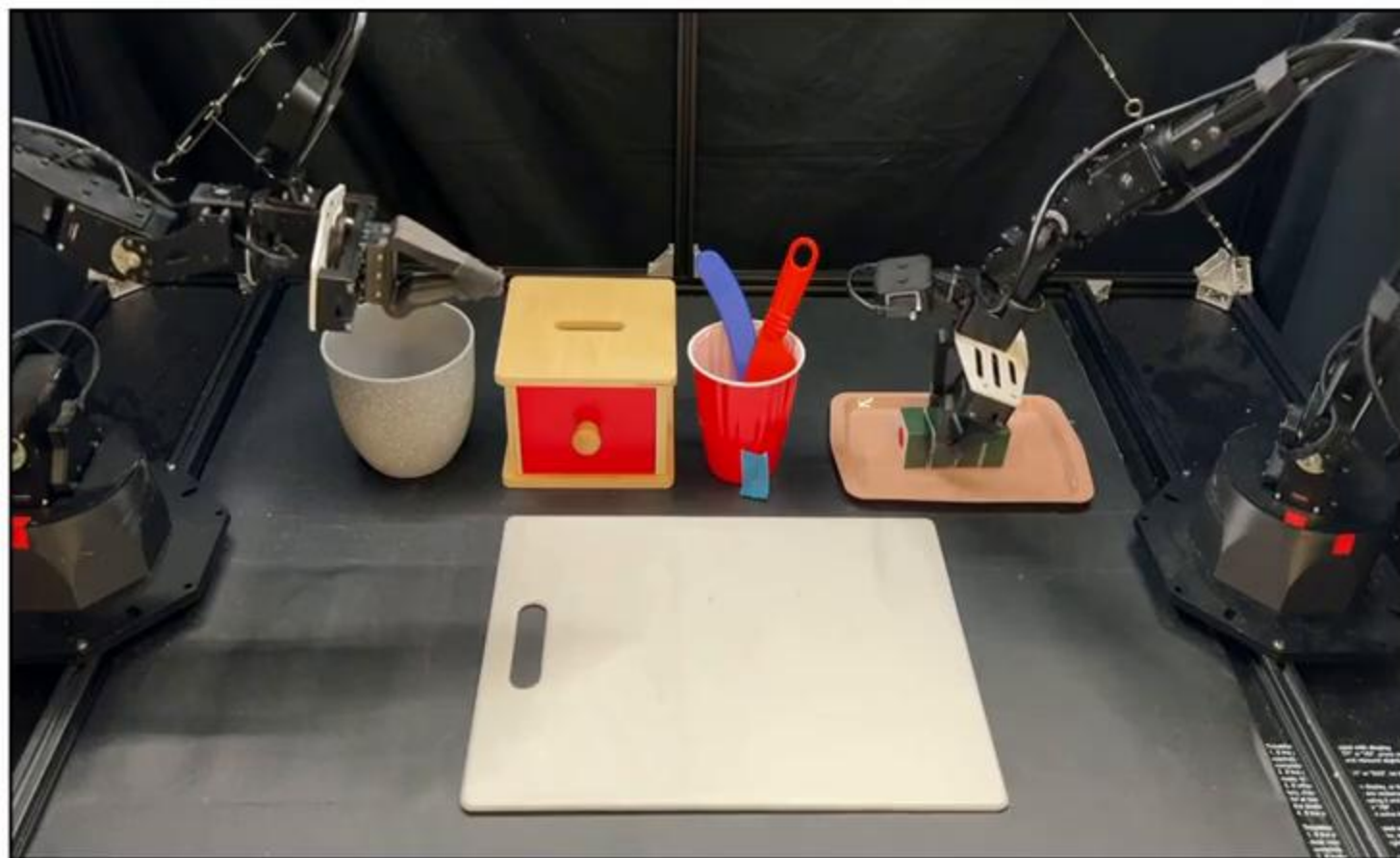


Perception  
Control  
...

Imitation Learning  
Reinforcement Learning  
Dynamics Learning  
Representation Learning  
...

# Sequential Decision Making

- What is the best sequence of actions to cut the sushi?

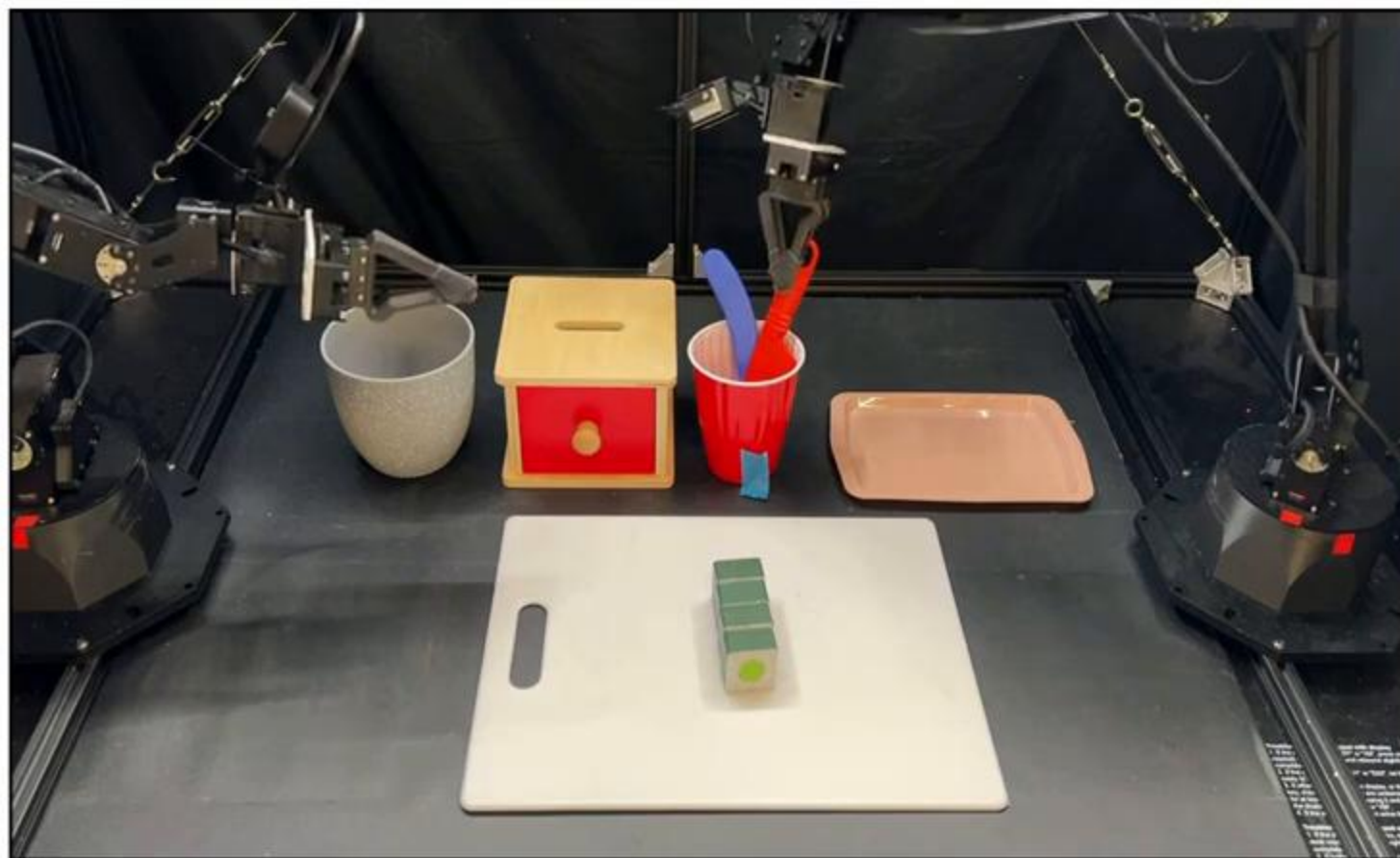


The Ingredients for Robotic Diffusion Transformers

Dasari, Mees et. al., ICRA 2024

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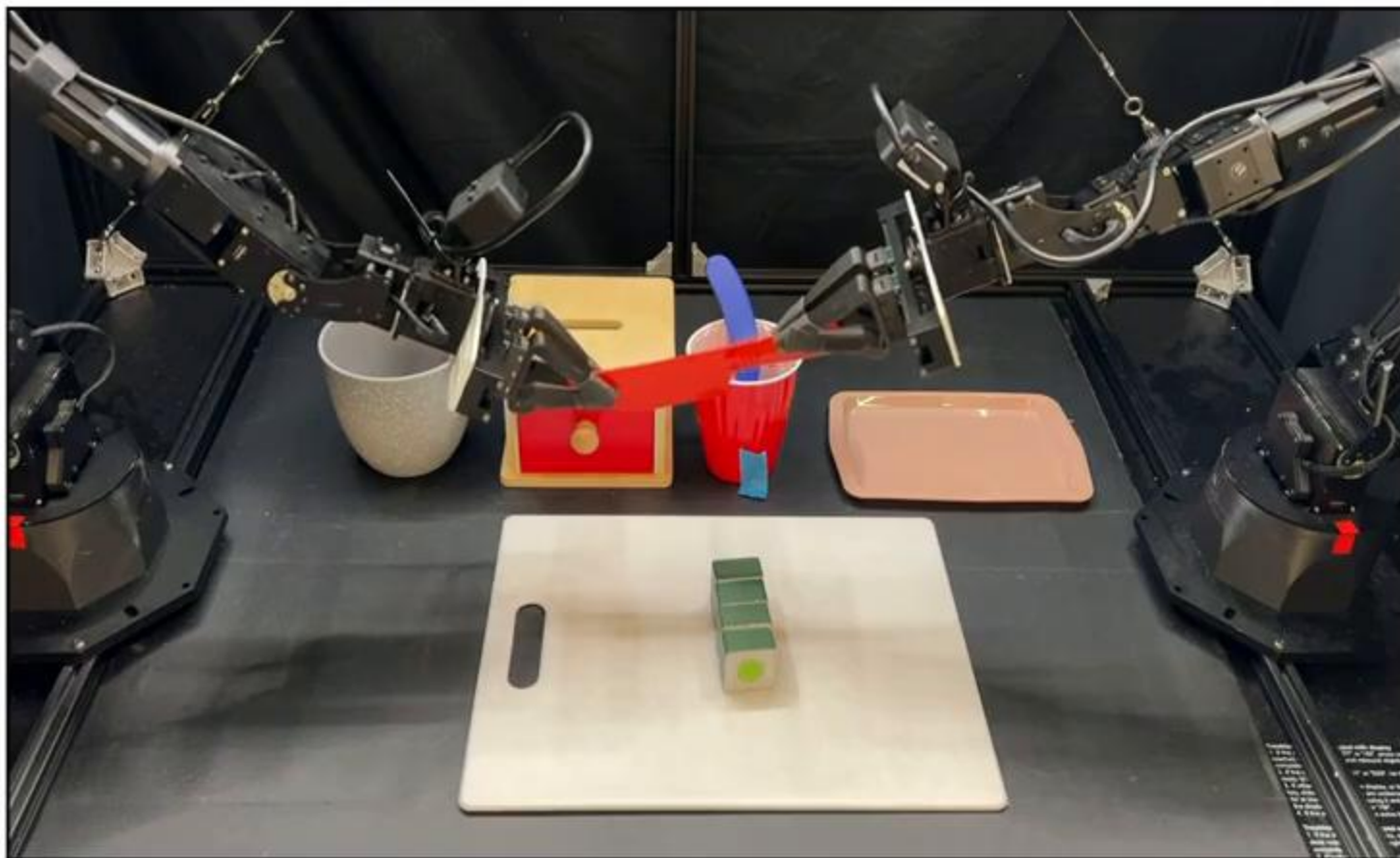


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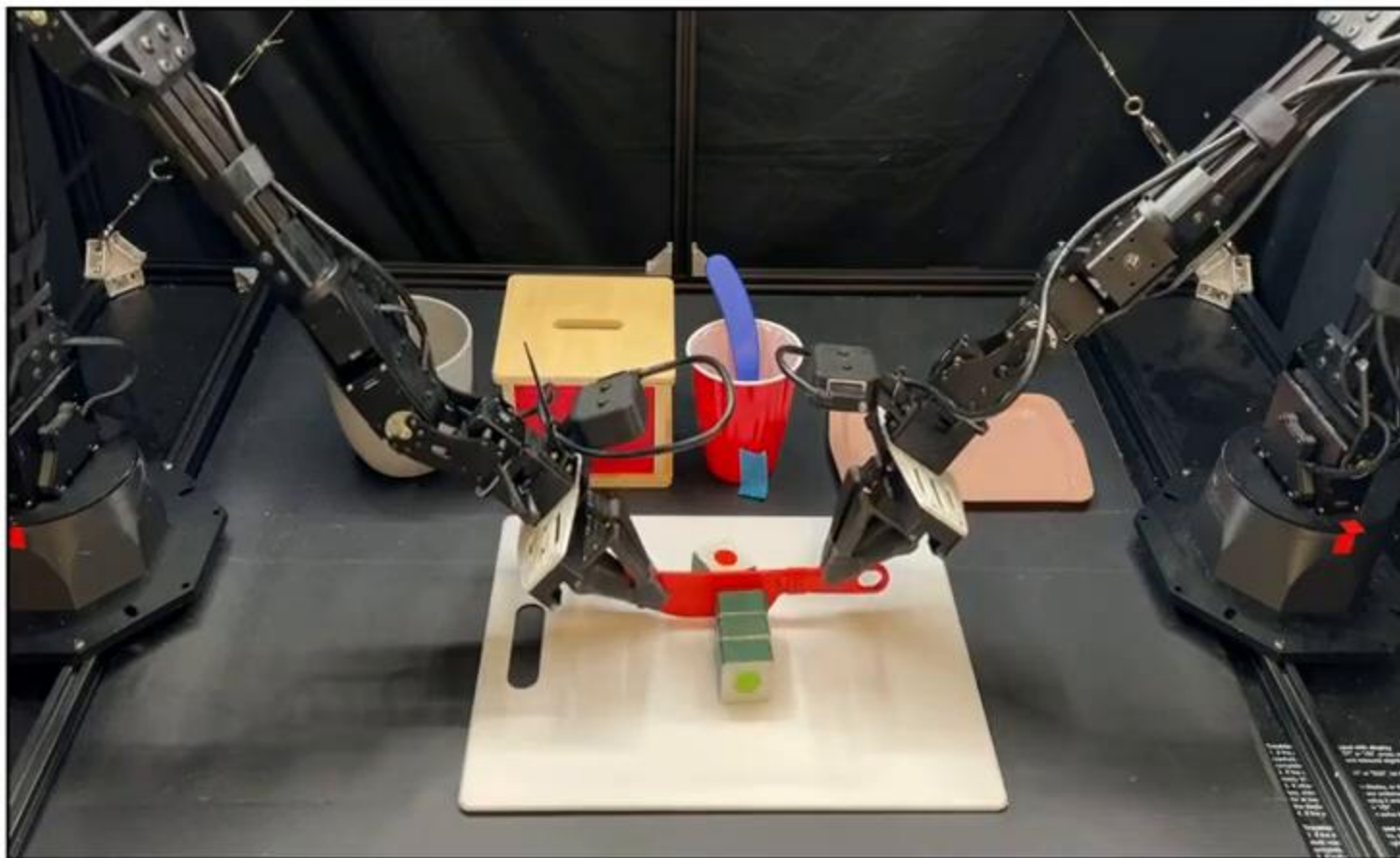


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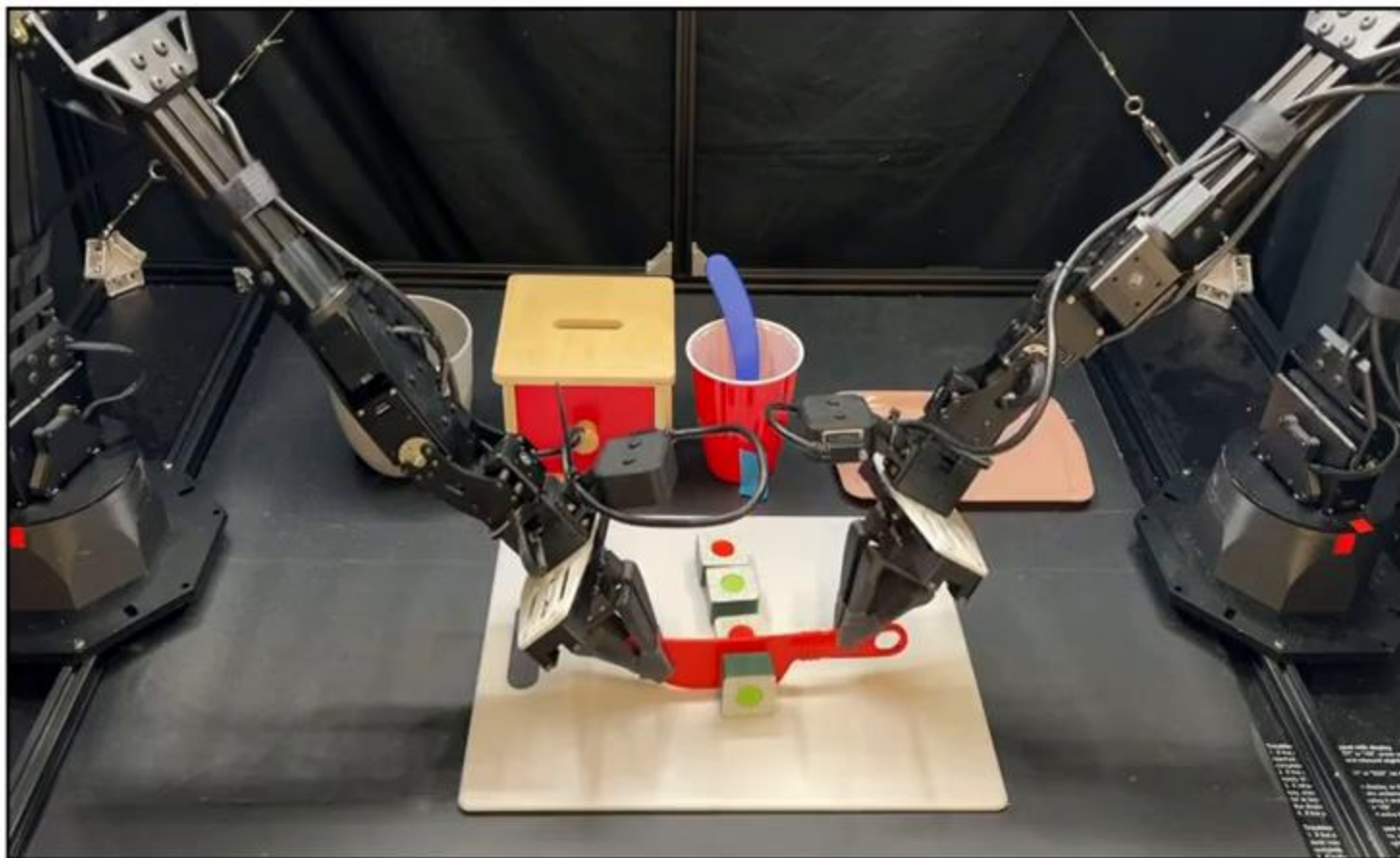


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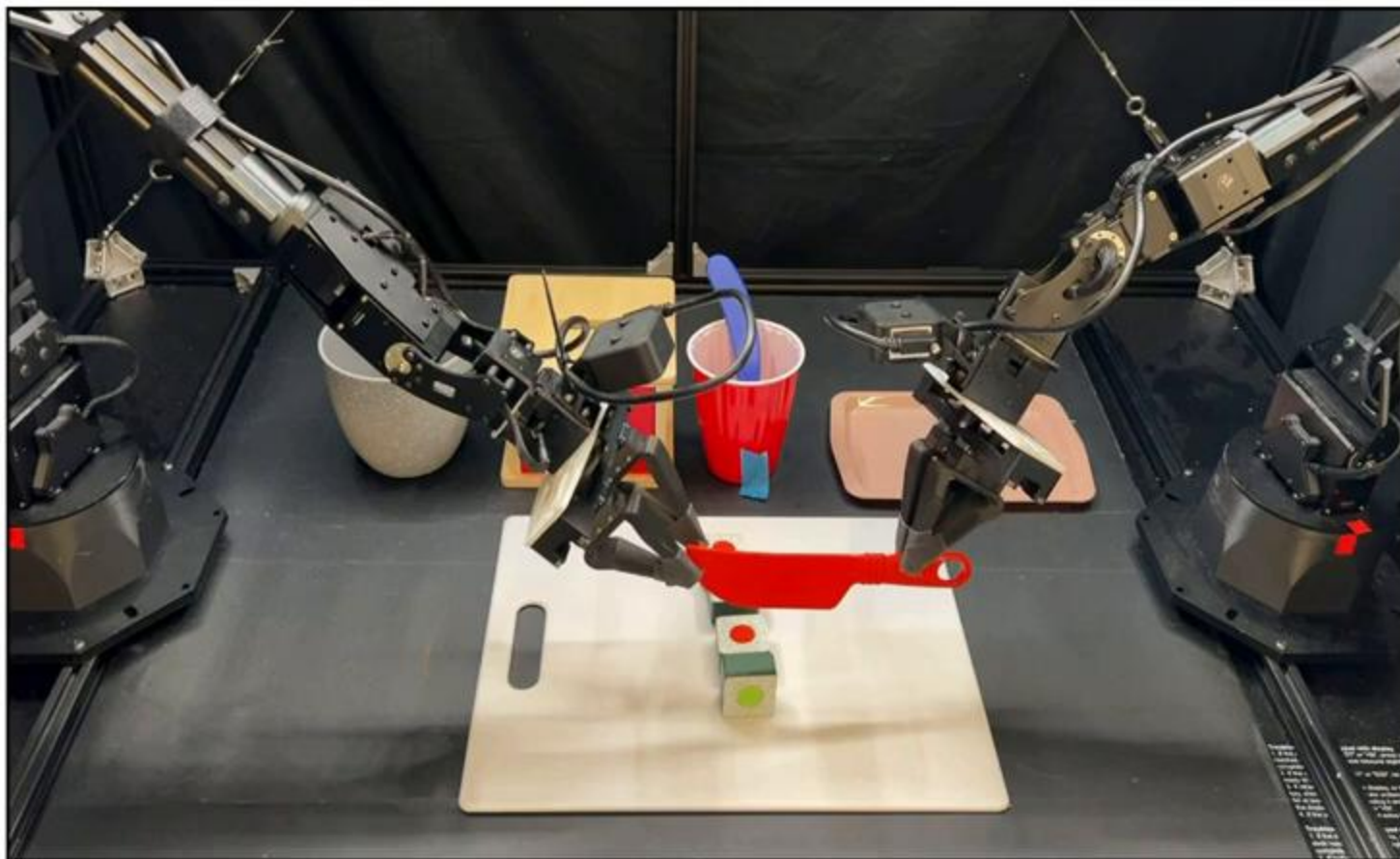


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Dasari, Mees et. al., ICRA 2024

# Sequential Decision Making

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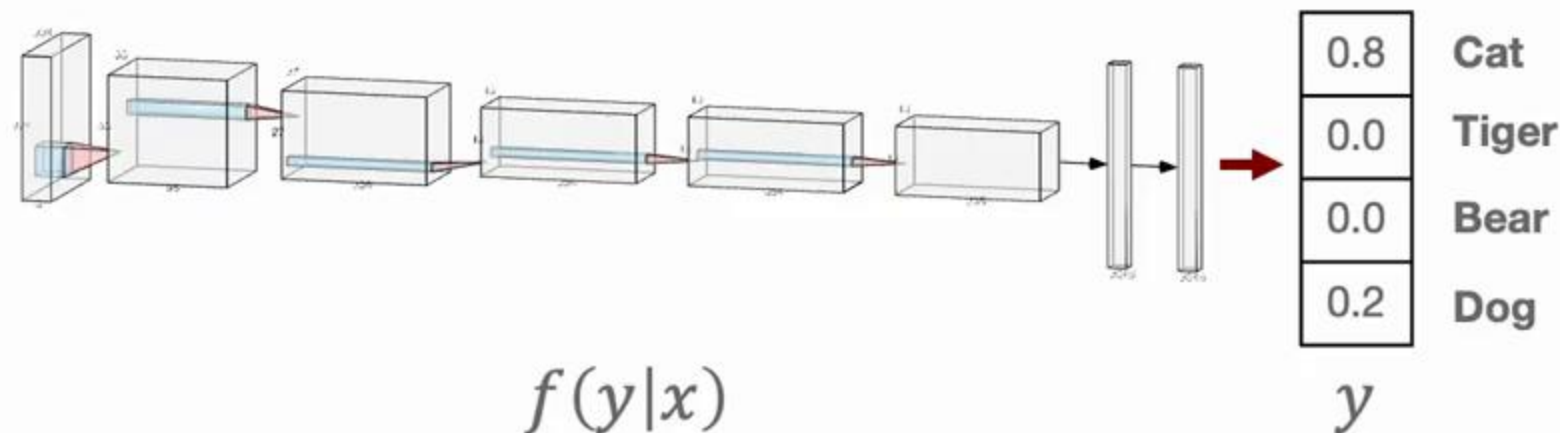
The Ingredients for Robotic Diffusion Transformers

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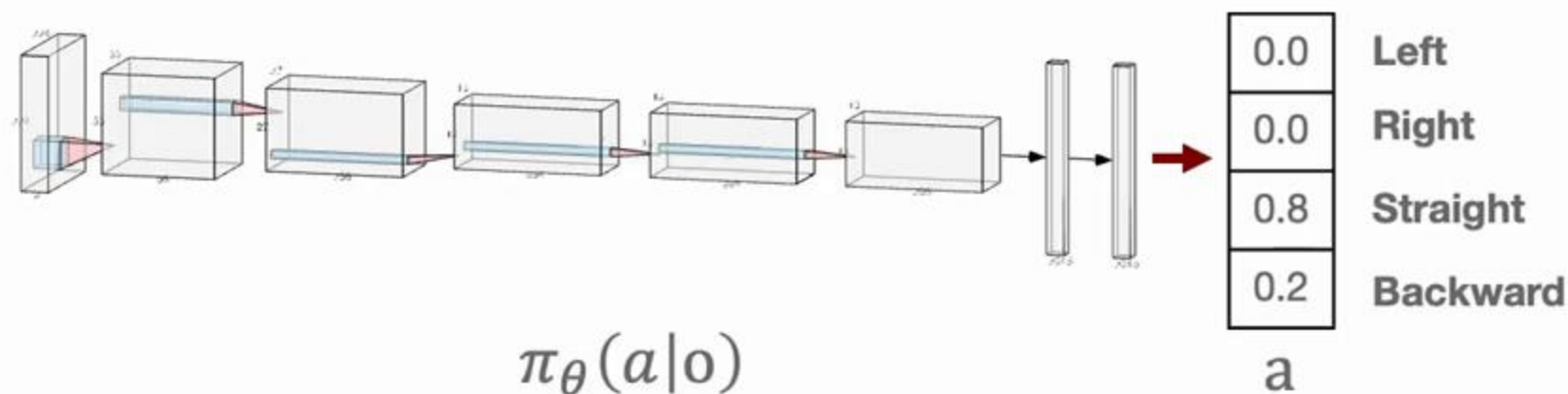
# From Pixels to Decisions: Representing a Policy



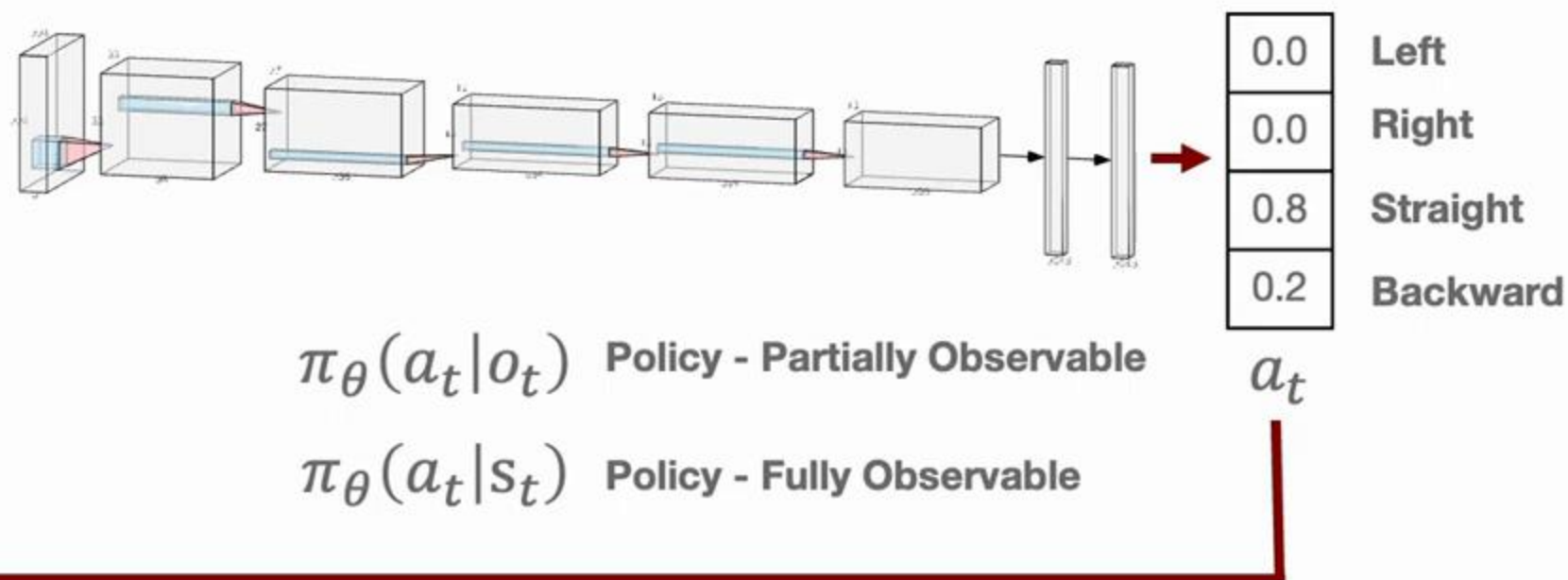
$x$



$o$



# Mapping Observations to Actions



$a_t$  - action

$o_t$  - observation

$s_t$  - state

# State vs Observation



$s_t$  - state



$o_t$  - observation

No matter the weather or visibility, the  
**state does not change!**

# State vs Observation



$s_t$  - state

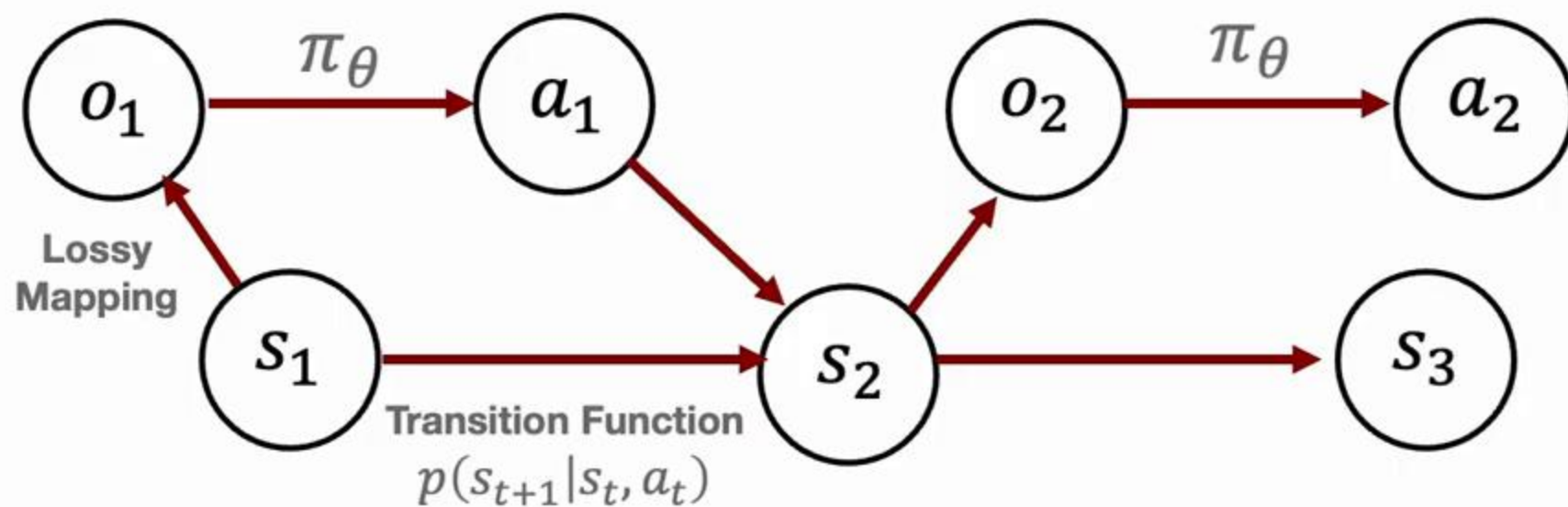


$o_t$  - observation

No matter the weather or visibility, the  
**state does not change!**

# Markov Property

- If you know  $S_2$  then  $S_1$  not necessary to determine  $S_3$



Andrey Markov

# Alternative Notations

$a_t$  - action

$s_t$  - state

$r(s, a)$  - reward

$$r(s, a) = -c(x, u)$$

$u_t$  - action

$x_t$  - state

$c(x, u)$  - cost function



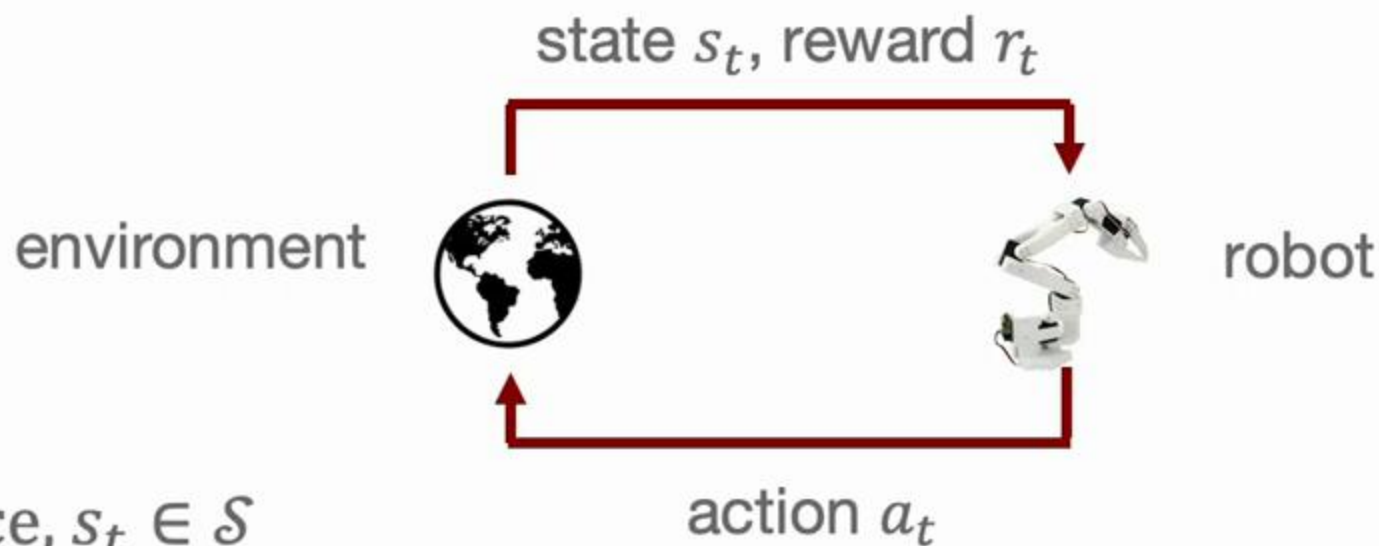
Richard Bellman



Lev Pontryagin

# Markov Decision Process

- Define by  $M = \langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R} \rangle$



- $\mathcal{S}$ : State space,  $s_t \in \mathcal{S}$
- $\mathcal{A}$ : Action space,  $a_t \in \mathcal{A}$
- $\mathcal{P}$ : Transition probability,  $s_{t+1} \sim \mathcal{P}(\cdot | s_t, a_t)$
- $\mathcal{R}$ : Reward function,  $r: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ ,  $r_t = R(s_t, a_t, s_{t+1})$

# MDP: State Space

Discrete



Continuous



# MDP: Action Space

Discrete



Continuous



# MDP: Transition Model

- Uncertainty caused by sensor noise, motor overshooting...

Deterministic

$$s_{t+1} = f(s_t, a_t)$$

Action: turn steering wheel 5°



Stochastic

$$s_{t+1} \sim \mathcal{P}(\cdot | s_t, a_t)$$

Action: turn steering wheel 5°



# MDP: Reward Function

## Sparse Rewards

e.g. 1 if grasped, 0 otherwise



## Dense Rewards

e.g. distance to object

Selected affordance  
region



Detected affordance  
region center



**Affordance Learning from Play for Sample-Efficient Policy Learning**

Borja, Mees et. al., ICRA 2022

# MDP: Reward Function

- How to detect if reward conditions are met?

## Sparse Rewards

e.g. 1 if grasped, 0 otherwise



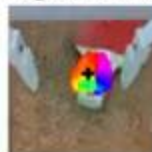
## Dense Rewards

e.g. distance to object

Selected affordance region



Detected affordance region center



**Affordance Learning from Play for Sample-Efficient Policy Learning**

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# Trajectory Probability

- Probability of a trajectory  $\tau = (s_0, a_0, s_1, a_1, \dots, s_T)$  under policy  $\pi$ :

- $$p_{\pi}(\tau) = \underbrace{\rho(s_0)}_{\text{Initial State Distribution}} \prod_{t=0}^{T-1} \underbrace{\pi(a_t | s_t)}_{\text{Policy}} \underbrace{P(s_{t+1} | s_t, a_t)}_{\text{Transition Function}}$$

Initial State  
Distribution

Policy

Transition Function



# Finite-Horizon vs Infinite-Horizon MDPs

- **Finite-Horizon:** the task has a strict time limit/steps  $H$
- **Infinite-Horizon:** interaction continues forever ( $T = \infty$ ) or until a "terminal state" is reached
  - **Discount Factor  $\gamma$ :** Rewards closer in time are higher weighted



# The Learning Objective

- Accumulated reward an agent receives:

$$G = \sum_{t=0}^{T-1} r_t \quad \text{or} \quad G = \sum_{t \geq 0} \gamma^t r_t$$

- Expected Return:

$$J(\pi) = \mathbb{E}_{\tau \sim \mathbb{P}_{\pi}(\tau)} [\sum_{t=0}^{T-1} r_t] \quad \text{or} \quad J(\pi) = \mathbb{E}_{\tau \sim \mathbb{P}_{\pi}(\tau)} [\sum_{t \geq 0} \gamma^t r_t]$$

- Optimal Policy:

$$\pi^* = \arg \max_{\pi} J(\pi)$$



**Thank you for your attention**